A LOCALLY FINITELY PRESENTED, FLAT, UNRAMIFIED MORPHISM OF ALGEBRAIC SPACES IS ETALE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a presentation reduction, which is the standard route for such lemmas in this chapter: the definition of etaleness for algebraic spaces is given through exactly this kind of presentation.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Etaleness of a morphism of algebraic spaces in terms of a source-and-target presentation. Properties of Spaces, Lemma spaces-properties-lemma-etale-local and Lemma lemma-local-source-target, used at proof lines 8078-8083. Produces the scheme morphism h and the reduction.
- The scheme case: locally finitely presented, flat and unramified implies etale. Morphisms, Lemma morphisms-lemma-flat-unramified-etale, used at proof lines 8083-8085. Gives that h is etale.
- Etaleness read off from the presentation. used at proof lines 8085-8085. Concludes for X -> Y.

Source reference. Original lemma 03MJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4751–4751. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3242–3242. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03FS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4503–4503. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *A locally finitely presented, flat, unramified morphism of algebraic spaces is étale.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $X \to Y$ be a locally finitely presented, flat, unramified morphism of algebraic spaces. By Properties of Spaces, Lemma 03FS we see this means there exists a diagram as in Lemma 03MJ with h locally finitely presented, flat, unramified and surjective vertical arrow a. By Morphisms, Lemma 02GV h is étale. Hence $X \to Y$ is étale by definition. □